

8051 Memory Addressing

1 Memory Organization Overview

Memory Division in 8051

The total memory in 8051 is logically divided into two types:

1. **Program Memory:** Stores programs to be executed (implemented using EPROM).
2. **Data Memory:** Stores data required for execution of programs (implemented using RAM).

Addressing Capability

The 8051 has a **16-bit Program Counter (PC)**, which means it can address up to:

$$2^{16} = 65536 \text{ bytes} = \mathbf{64 \text{ KB}}$$

2 Program Memory

Program Memory

Program memory stores the instructions (code) to be executed by the microcontroller. It is implemented using **EPROM** (Erasable Programmable Read-Only Memory) or Flash memory.

2.1 Program Memory Categories

Category	Size	Address Range	Description
On-chip (Internal)	4 KB	0000H – 0FFFH	Physically exists on the chip
External	64 KB	0000H – FFFFH	Externally interfaced memory

Address Overlap

The address map of **internal memory overlaps with external memory**. Both start from 0000H. They are distinguished using the **PSEN (Program Store Enable)** signal:

- **PSEN = 0:** Accessing external program memory.
- **PSEN = 1:** Not accessing external program memory.

2.2 Control Signals

EA and PSEN Signals

EA (External Access) Pin:

- **EA = 1 (HIGH):** CPU fetches code from internal ROM (0000H–0FFFH) first, then external memory.
- **EA = 0 (LOW):** CPU fetches all code from external program memory only.

PSEN (Program Store Enable):

- Active LOW signal used as read strobe for external program memory.
- Generated during instruction fetch cycles from external memory.

3 Data Memory

Data Memory

Data memory stores variables, temporary data, and the stack during program execution. It is implemented using **RAM** (Random Access Memory).

3.1 Data Memory Categories

3.1.1 On-chip (Internal) Data Memory

Internal data memory consists of **two parts**:

Component	Address Range	Description
RAM Block	00H – 7FH (128 bytes)	General-purpose data storage
Special Function Registers	80H – FFH	Addresses allotted to SFRs

3.1.2 External Data Memory

- **Size:** 64 KB can be addressed (0000H – FFFFH).
- **Addressing:** Register **DPTR** stores addresses for memory access.
- **Control Signals:** 8051 generates **RD** and **WR** signals during memory access.
- **Chip Select:** Derived from address lines for selecting external memory chips.

4 Internal RAM Addressing Modes

Addressing Modes for Internal Memory

128-byte RAM (00H – 7FH):

- Can be accessed using **direct** or **indirect** addressing mode.

Special Function Registers (80H – FFH):

- Accessible **only with direct** addressing mode.

4.1 256-Byte Version (8052)

Extended RAM in 8052

The 8052 variant has **256 bytes** of internal RAM (00H – FFH):

Memory Region	Address	Addressing Mode
Lower 128 bytes	00H – 7FH	Direct or Indirect
Upper 128 bytes	80H – FFH	Indirect only
SFR Bank	80H – FFH	Direct only

Address Overlap in 8052

In the 8052, the **SFR addresses (80H–FFH)** overlap with upper 128 bytes RAM. They are distinguished by the addressing mode used:

- **Direct addressing** → Accesses **SFRs**.
- **Indirect addressing** → Accesses **upper 128 bytes RAM**.

5 Internal RAM Memory Map

RAM Block Organization

The lower 128 bytes of internal RAM (00H – 7FH) are functionally organized into **3 sections**:

Address Range	Section	Function
00H – 1FH	Register Banks	Working registers (R0–R7)
20H – 2FH	Bit-Addressable RAM	128 individually addressable bits
30H – 7FH	General-Purpose RAM	Byte-addressable storage, stack

6 Section 1: Register Banks (00H – 1FH)

Register Banks

The **lowest 32 bytes** (00H – 1FH) are divided into **4 register banks**, each containing **eight 8-bit registers** (R0–R7).

Bank	RS1:RS0	Address Range	Registers
Bank 0	00	00H – 07H	R0 – R7
Bank 1	01	08H – 0FH	R0 – R7
Bank 2	10	10H – 17H	R0 – R7
Bank 3	11	18H – 1FH	R0 – R7

Bank Selection and Stack

Default Bank:

- **Bank 0** (00H–07H) is the **reset bank** (selected after reset).
- Bank selection bits **RS1** and **RS0** are present in **PSW register**.

Stack Pointer Initialization:

- Stack Pointer (SP) is initialized to **07H** after reset.
- First PUSH stores data at **08H** onwards (Bank 1 area).
- If using multiple banks, relocate SP to avoid conflicts (e.g., SP = 2FH or 7FH).

7 Section 2: Bit-Addressable RAM (20H – 2FH)

Bit-Addressable Memory

The second section extends from **20H to 2FH** (16 bytes), forming a **bit-addressable block** of memory:

$$16 \text{ bytes} \times 8 \text{ bits/byte} = \mathbf{128 \text{ bits}}$$

Each bit can be individually addressed from **00H to 7FH** (**bit addresses**).

7.1 Bit Address Mapping

Byte Address	Bit Addresses	Alternative Notation
20H	00H – 07H	20.0 – 20.7
21H	08H – 0FH	21.0 – 21.7
22H	10H – 17H	22.0 – 22.7
...	...	...
2FH	78H – 7FH	2F.0 – 2F.7

Two Ways to Access Bits

Method 1: Direct bit address

```
SETB 05H      ; Set bit 5 of byte 20H
CLR  10H      ; Clear bit 0 of byte 22H
```

Method 2: Byte.Bit notation

```
SETB 20H.5    ; Set bit 5 of byte 20H (same as 05H)
CLR  22H.0    ; Clear bit 0 of byte 22H (same as 10H)
```

8 Section 3: General-Purpose RAM (30H – 7FH)

General-Purpose RAM

The third block occupies addresses from **30H to 7FH** (80 bytes). This is **byte-addressable memory space** used for:

- General data storage.
- **Stack memory** (when SP is relocated here).
- Temporary variables and buffers.

Internal Memory Access

All internal data memory locations (00H – 7FH) are accessed using **8-bit addresses** under appropriate modes of addressing:

- **Direct:** Address specified directly in instruction.
- **Indirect:** Address held in R0 or R1 register (@R0, @R1).

9 Complete Memory Map Diagram

8051 Internal RAM Memory Map

Address	Content	Access Mode
Special Function Registers		
80H – FFH	SFRs (ACC, B, PSW, SP, DPTR, Ports, etc.)	Direct only
Internal RAM (128 bytes)		
30H – 7FH	General-purpose RAM / Stack	Direct or Indirect
20H – 2FH	Bit-addressable RAM (128 bits)	Direct or Indirect
18H – 1FH	Register Bank 3 (R0–R7)	Direct or Indirect
10H – 17H	Register Bank 2 (R0–R7)	Direct or Indirect
08H – 0FH	Register Bank 1 (R0–R7) / Default Stack	Direct or Indirect
00H – 07H	Register Bank 0 (R0–R7) – Default	Direct or Indirect

10 Summary: Memory Addressing

Quick Reference Summary

Program Memory:

- Internal: 4 KB (0000H–0FFFH), External: 64 KB (0000H–FFFFH).
- Distinguished by **PSEN** signal; controlled by **EA** pin.

Data Memory:

- Internal RAM: 128 bytes (00H–7FH) + SFRs (80H–FFH).
- External: 64 KB via DPTR with RD/WR signals.

Internal RAM Sections:

- **00H–1FH**: 4 Register Banks (R0–R7 each).
- **20H–2FH**: Bit-addressable (128 bits, addresses 00H–7FH).
- **30H–7FH**: General-purpose / Stack memory.

Addressing Modes:

- Lower 128 bytes: Direct or Indirect.
- SFRs: Direct only.
- Upper 128 bytes (8052): Indirect only.

Understanding memory organization is crucial for efficient 8051 programming and resource management.